Distance, Capacity and Inequality in Access to Government and Aided Schools: A Spatial Big Data Analysis of India and Kerala

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Abstract

This study measures how far the children in India live from a Government or Aided school, and how many school places are available for them near their home. For this, 140 million populated 100 m population cells from WorldPop, 1,070,017 Government and Aided schools from UDISE, and the official distance norms of 1, 3, 5 and 7 km are used. Access is measured with the help of the Enhanced Two-Step Floating Catchment Area (E2SFCA) method, which is run on Apache Spark along with Apache Sedona. Kerala is studied in more detail with its own school data. About 12.4 million primary children in India (16 %) live more than 1 km away from a primary school, and higher secondary has only 0.40 places per child. The inequality is high, with a Gini index of 0.60 for primary places per child, and most of this inequality lies inside the states, not between them. In Kerala, 19 % of the lower primary children are beyond 1 km, mostly in Idukki and Wayanad. The results are shown in an offline map dashboard. A benchmark showed that: Spark becomes faster than plain Python above about one million cells, and it also made the full India run possible on a single laptop.

1. Introduction

A school is useful to a child only if it is close enough to reach and also has enough room to take the child. According to the Right to Education (RTE) Model Rules (Government of India, 2010), a primary school should be within a walking distance of 1 km, and an upper primary school should be within 3 km. Samagra Shiksha (Ministry of Education, 2022) aims for a secondary school within 5 km (3 km in hilly areas) and a higher secondary school within 7 to 10 km. In this study, 7 km is used for higher secondary schools. Most of the government reports count the number of schools per state or per district. This hides the village which is 4 km away from its nearest school, and it also hides the school which is close but already full. Hence distance alone is not enough. A child who lives near a crowded school and a child who lives far from an empty school both have poor access, but for different reasons. Because of this, two things were measured for every 100 m cell in India, i.e. the distance to the nearest school of each level, and the number of school places per child within reach. The question is simple. How many children live beyond the distance norms, how many school places are available per child, and how unequal this is across states, districts and local bodies. Only Government and Aided schools are considered here, because these are the schools which are provided by the state. Kerala is taken as a case study, because its school data is the most complete and it could be checked against the official enrolment. This is a big data problem. The WorldPop grid for India has 1.21 billion cells, and the access step needs every pair of a cell and a school within 7 km. The first plan also included health facilities for India and fire and rescue stations for Kerala, but their capacity data could not be obtained in time. Hence this report covers only schools (see 4.3).. The cell-school pairs cannot be computed in a normal GIS desktop tool, so I used Apache Spark (Zaharia et al., 2016) with Apache Sedona, which grew out of GeoSpark (Yu et al., 2015).

2. Data and methods

2.1 Study area and base data

The study area is all of India, with Kerala as a detailed case study. Area units came from geoBoundaries (Runfola et al., 2020): 36 states and union territories, 735 districts and 6,824 sub-districts. For Kerala I also used 1,027 local body boundaries (grama panchayats, municipalities and corporations) from OpenStreetMap (OpenStreetMap contributors, 2026).Officially there are 1,034, hence 7 of them are missing in OpenStreetMap. These are shown as grey patches on the maps.

The distance norms which are used for each school level are given below. For Kerala, its own school levels are used, which divide the classes in a slightly different way.

Level	India classes	Kerala classes	Norm	Source
Primary / Lower primary	1–5	1–4	1 km	RTE Model Rules 2010
Upper primary	6–8	5–7	3 km	RTE Model Rules 2010
Secondary / High school	9–10	8–10	5 km (3 km hilly)	Samagra Shiksha
Higher secondary	11–12	11–12	7 km	Samagra Shiksha

2.2 Population and school age children

The population data is taken from WorldPop (2025), using the 2026 estimate at 100 m resolution (constrained, CC BY 4.0). According to this data, there are 1,470.3 million people in India. WorldPop gives the ages in 5-year bands, hence each band is split evenly into single years. Eg: the primary age group 6–10 is taken as 0.8 of the 5–9 band plus 0.2 of the 10–14 band. Not every child goes to a Government or Aided school. Hence the demand in each cell is multiplied by the share of pupils in these schools, which is computed from UDISE for each state and level. In India this share is about 65 % for primary. In Kerala it is 61 % for lower primary and 88 % for higher secondary. For Kerala, the estimated demand was within 4 % of the real 2026-27 enrolment.

2.3 Schools and their capacity

The India school list is the UDISE school-wise file from the India Data Portal (2022), the only open school-level file with location, management, pupils and teachers. It was taken from the UDISE+ school search on 12 January 2022, so it most likely shows the 2020-21 school year. After keeping only Government and Aided schools, and taking the Kerala rows from the verified Kerala layer, the India list has 1,070,017 schools. For Kerala I used the verified school layer of 11,764 schools with 2026-27 pupils and teachers from the Sametham portal (KITE, 2026), plus higher secondary seats from HSCAP (Government of Kerala, 2026a), at 50 per batch, and VHSE seats from VHSCAP (Government of Kerala, 2026b), at 30 per batch.

No source publishes how many seats a school has. So the main capacity was based on norms: the number of teachers at a level multiplied by the pupil-teacher ratio of the RTE and Samagra rules (30 for primary, 35 for upper primary, 30 for secondary and higher secondary). The current enrolment was used as a second, sensitivity run.

2.4 Making the data consistent

Many UDISE coordinates are not the school building. Some are placeholders, with four or more schools sharing one point. These were moved to an OpenStreetMap school of the same name within 3 km, or else to the median of good schools in the same village. On Kerala, where true locations are known, this cut the median location error from 104 m to 80 m.

I ran 107 data quality checks. Two states have incomplete school lists in UDISE: Delhi has only about 40 % of the expected pupils, and Andhra Pradesh is missing its junior colleges for higher secondary. Dadra and Nagar Haveli and Daman and Diu has no UDISE rows. These state-levels are kept on the map but flagged, and the national indices are given with and without them. The cells which are near a state border and were given a district of the neighbouring state are moved to the nearest district of their own state.

2.5 Big data processing

All data was stored as Parquet and GeoParquet files in a data lake with three layers: raw (bronze), cleaned (silver) and results (gold). Spark read the WorldPop raster in blocks inside its tasks and kept only cells with people. This gave a total of 140,086,378 populated cells, and their total matches the raster sum. Each cell is assigned to the area which contains its centre point.

The main step is the E2SFCA method (Luo and Qi, 2009), which improves the two-step floating catchment area method (Luo and Wang, 2003) by adding distance decay.In step 1, each school is given a ratio of its places to the number of children within its catchment. In step 2, each cell adds up the ratios of all the schools which it can reach. In both the steps, the children and the schools are weighted with the help of a Gaussian distance decay (Dai, 2010), where d is the distance and d0 is the catchment size:

$$f(d) = [\, \exp(-\tfrac{1}{2}\,(d / d_0)^2) - \exp(-\tfrac{1}{2}) \,] \,/\, [\, 1 - \exp(-\tfrac{1}{2}) \,], \quad \text{for } d \le d_0 \text{ (and 0 beyond } d_0)$$

The result, A, is known as the number of school places per child. A value of 1 means that there is one place for every child. The distances are measured as geodesic distances on the WGS84 ellipsoid (Sedona ST_DistanceSpheroid).

Before the main run, four ways of doing the cell-school join in Sedona were compared. The broadcast index join, which sends the small school table to every worker, was found to be the fastest. Eg: for Uttar Pradesh it took 25 s, whereas the Quad-tree join took 72 s and the H3 join took 103 s, and the KDB-tree join did not finish at all. The India run was done state by state, so that it could restart after a failure. It took about 5 hours and wrote a result of 17.7 GB. For every state, the sum of A times children was equal to the sum of the school places. This is the check which is guaranteed by the method itself.

2.6 Analysis and decision support

The cell results are added up for the states, districts, sub-districts and Kerala local bodies, weighted by the number of children. For each unit and level, three things are reported, i.e. the share of children beyond the norm, the mean distance to the nearest school and the mean places per child.

Inequality was measured with the Gini index and the Theil index (Theil, 1967). Gini gives one number from 0 (equal) to 1. Theil can be split exactly into a part between states and a part within states. Hot and cold spots were found with the Getis-Ord Gi* statistic (Getis and Ord, 1992) in Sedona, with a 36 km distance band for sub-districts and 11 km for Kerala local bodies, and a 5 % false discovery rate (Benjamini and Hochberg, 1995). Sub-districts were grouped into area types with K-Means in Spark MLlib, using 12 standardised indicators (three measures for each of the four levels). K = 5 had the best silhouette score (Rousseeuw, 1987), 0.415. FP-Growth

(Han et al., 2000) found which access problems tend to occur together. In Kerala, neighbouring cells beyond the norm were joined into clusters, which act as priority areas for new schools.

All of this is shown in an offline web dashboard, a single 12.7 MB HTML file built with Leaflet (Agafonkin, 2023). It has a map for every state and union territory, Kerala down to local bodies, and panels for the inequality numbers, area types and rules.

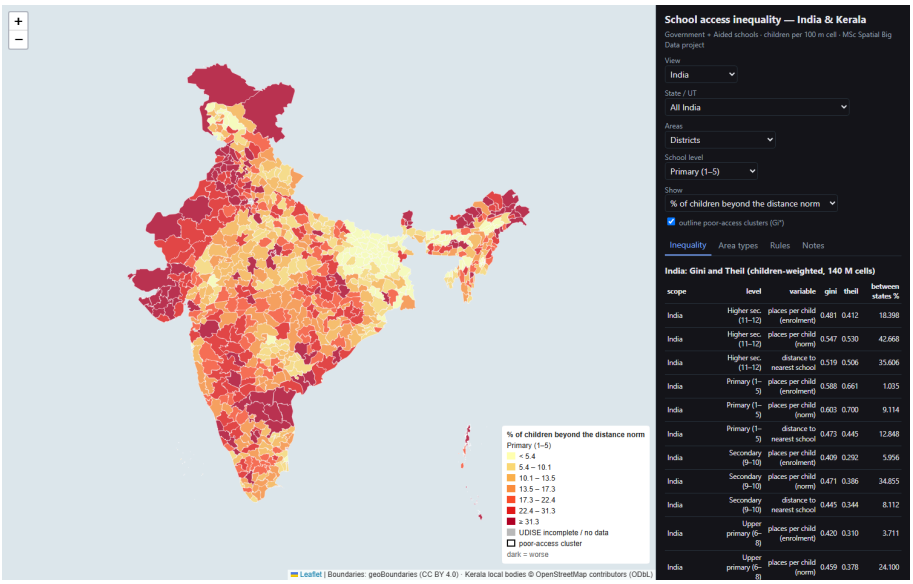


Figure 1. The offline dashboard, India view: districts coloured by the share of primary children beyond 1 km, with the inequality panel on the right.

3. Results

3.1 Access to schools in India

Distance is mainly a primary school problem, while capacity is mainly a higher secondary problem. About 12.4 million primary children live more than 1 km from a primary school. At higher secondary there are only 0.40 places per child, so near their home there is a Government or Aided seat for only about 4 in 10 children of that age. The children counted here are those expected to use Government and Aided schools.

Level	Children (million)	Beyond norm	Children beyond norm (million)	Mean distance to nearest school	Places per child (norm)	Places per child (enrolment)
Primary (1 km)	77.4	16.0 %	12.4	0.64 km	1.10	0.92
Upper primary (3 km)	50.3	3.0 %	1.5	0.96 km	1.01	0.82
Secondary (5 km)	32.3	3.6 %	1.2	1.74 km	0.71	0.73
Higher secondary (7 km)	31.9	9.7 %	3.1	3.35 km	0.40	0.46

The states differ a lot. Bihar has the fewest places at every level: 0.56 per child at primary and 0.13 at higher secondary. Jharkhand and Uttar Pradesh come next. For primary distance, the worst states are Punjab (39 % of children beyond 1 km), Andhra Pradesh (34 %), Arunachal Pradesh (32 %) and Gujarat (28 %). For higher secondary distance, Karnataka (26 %) and Odisha (25 %) are the worst. The two capacity runs give a similar picture: the district rankings have a Spearman correlation of 0.64 to 0.70, and 6 or 7 of the 10 worst districts are the same in both. Figure 2 shows why both measures are needed. Bihar is light in panel (a), because only 2.8 % of its primary children live beyond 1 km, but it is the darkest in panel (b), because its schools are crowded, with 0.56 places per child.

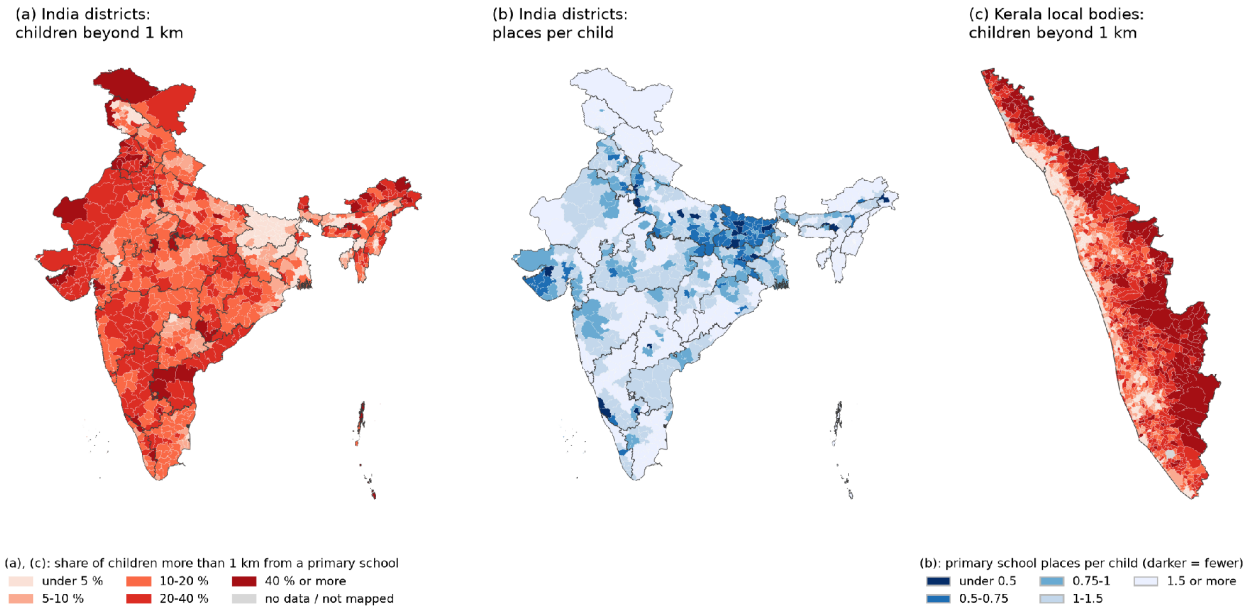


Figure 2. Primary level only (1 km norm). (a) India districts: share of children living more than 1 km from the nearest primary school (classes 1–5). (b) India districts: primary school places per child from E2SFCA, where darker means fewer places. (c) Kerala local bodies: share of lower primary children (classes 1–4) beyond 1 km. The other levels are given in the tables and in the dashboard. Grey areas have incomplete UDISE data (Delhi, Dadra and Nagar Haveli and Daman and Diu) or no OpenStreetMap local body.

3.2 Kerala case study

Kerala has schools close to almost every child above lower primary, but lower primary distance is still a problem in the hills. Overall 19.3 % of lower primary children live more than 1 km from a lower primary school. In Idukki it is 44 % and in Wayanad 42 %, against 8.5 % in Kozhikode. Above lower primary, less than 1 % of children are beyond the norm in Kerala as a whole, and the highest district value is 6.6 % (upper primary in Idukki).

Level	Children (million)	Beyond norm	Mean distance	Places per child (norm)	Places per child (enrolment)
Lower primary (1 km)	1.00	19.3 %	0.69 km	1.32	0.97
Upper primary (3 km)	0.96	1.0 %	0.91 km	0.99	0.97
High school (5 km, 3 km hilly)	1.11	0.5 %	1.29 km	1.36	1.03

Level	Children (million)	Beyond norm	Mean distance	Places per child (norm)	Places per child (enrolment)
Higher secondary (7 km)	0.84	0.1 %	1.52 km	0.82	0.95

The groups of neighbouring cells beyond the norm gave 411 priority areas with at least 50 children each. 360 of them are for lower primary and hold about 171,000 children. The largest is centred in Vadavukode-Puthencruz in Ernakulam (about 26,000 children), followed by Chalakudy municipality in Thrissur (about 17,000).

3.3 Inequality and patterns

Places per child are very unequal, and most of the inequality is inside states. For primary, only 9 % of the Theil index is between states. The other 91 % is between districts and villages of the same state. The between-state share grows with the level and reaches 43 % at higher secondary, where whole states like Bihar and Uttar Pradesh fall short.

Level	Gini, places per child	Gini, distance	Theil share between states
Primary	0.60	0.47	9 %
Upper primary	0.46	0.45	24 %
Secondary	0.47	0.44	35 %
Higher secondary	0.55	0.52	43 %

Leaving out the flagged states with incomplete data changes these Gini values by 0.02 at most. In Kerala the Gini for lower primary places is 0.61, and 29 % of the Theil index lies between local bodies.

The Gi* test found 952 sub-districts in significant hot spots of primary children beyond the norm, and 606 in cold spots of primary places per child. K-Means gave five area types, named from their average values. The largest is "crowded: near but few places" (2,737 sub-districts, 53.5 million primary children), where schools are only 0.5 km away but there are 0.90 primary places per child and 0.31 at higher secondary. The others are "well served" (1,882 sub-districts, 2.1 primary places per child), "higher secondary far" (796, with 39 % of children beyond 7 km), "far from primary school" (415, with 44 % beyond 1 km) and "remote: far at every level" (60, with 84 % beyond 1 km and 10 km to the nearest higher secondary school).

The biggest group by far is the crowded one, mostly in Bihar, Madhya Pradesh and West Bengal. FP-Growth showed that shortages come together. A sub-district with few places at primary, upper primary and higher secondary also has few places at secondary in 96 % of cases (lift 4.0).

3.4 Big data processing performance

I timed the core E2SFCA step for primary schools (every cell-school pair within 1 km, the decay weight, and the sum per school) on three data sizes. Each setting ran in its own process, three times, and the median is reported. Both engines gave exactly the same pairs and the same weighted totals, for example 27,744,351 pairs for Uttar Pradesh. A brute-force count on 2,000 random cells also matched.

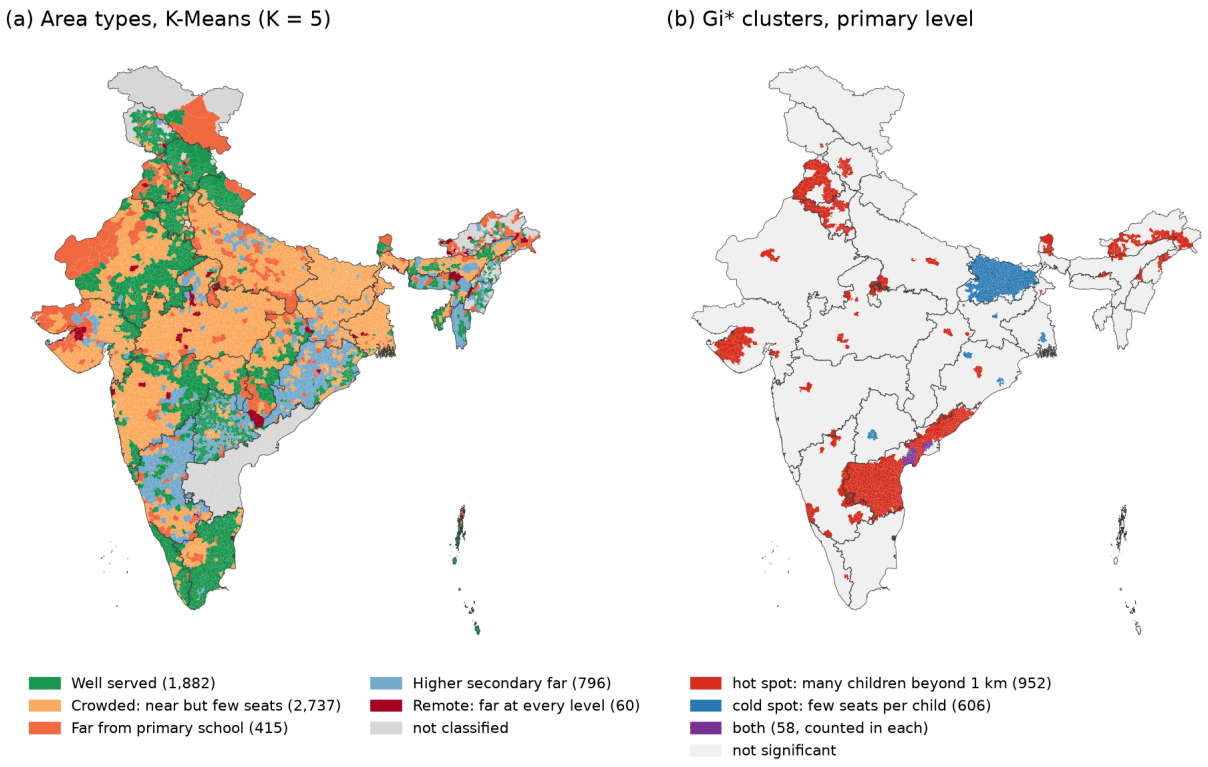


Figure 3. (a) Five area types from K-Means for 5,890 India sub-districts; grey areas are not classified (incomplete school data or fewer than 100 children). (b) Gi clusters at primary level, i.e. hot spots of children beyond 1 km (952 sub-districts) and cold spots of few seats per child (606); 58 are both.*

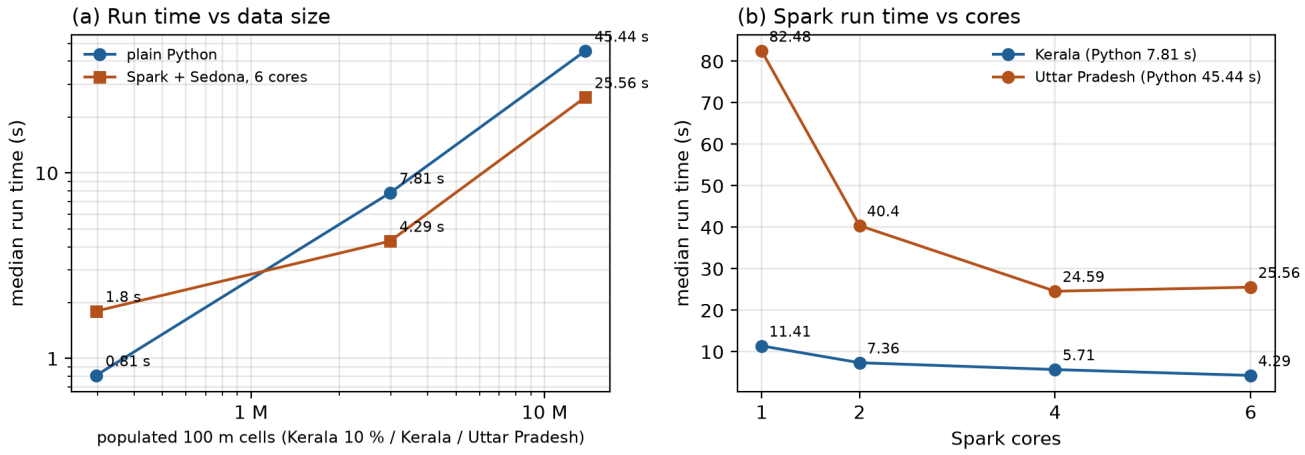


Figure 4. Benchmark of the core E2SFCA step (primary, 1 km), median of 3 runs. (a) Plain Python vs Spark + Sedona on 6 cores for three data sizes. (b) Spark run time on 1, 2, 4 and 6 cores for Kerala and Uttar Pradesh

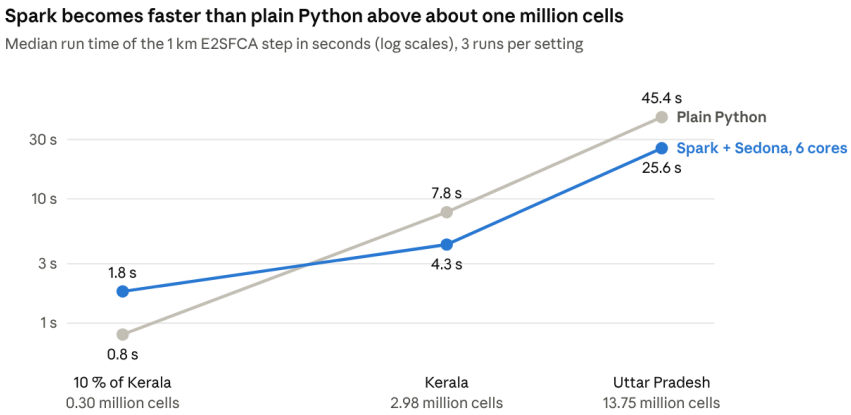


Figure 3. Run time against data size. Source: notebook 30, benchmarks.csv, 11 November 2026

On the small sample plain Python is faster, because Spark spends time planning and scheduling tasks. Spark also needs 7.5 to 10 s to start, which is not included above. From the size of Kerala upwards, Spark on 6 cores is about 1.8 times faster. Plain Python did not run out of memory (2.8 GB for Uttar Pradesh), because it works in chunks of one million cells.

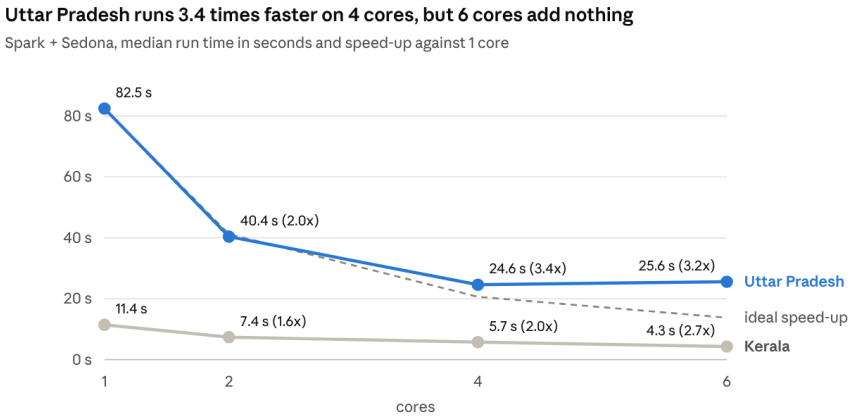


Figure 4. Spark run time against cores. Source: notebook 30, benchmarks.csv, 11 November 2026

Adding cores helped up to 4. Uttar Pradesh ran 2.0 times faster on 2 cores and 3.4 times faster on 4, but 6 cores gave nothing more. The laptop CPU (Intel Core 5 210H) has only 4 fast performance cores and 4 slower efficiency cores, so the fifth and sixth tasks land on slow cores.

4. Discussion

4.1 What the results mean

There are two different access problems in India, and they need different answers. The first is distance. It affects primary children most, in hill and forest areas and in thinly settled districts, and the answer is a new school or school transport. The second is capacity. In the large crowded group of sub-districts, schools are close but there are too few places and teachers, so the answer is more classrooms and teachers in existing schools, not more buildings.

The Theil split shows where to plan.For primary and upper primary, almost all the inequality lies inside the states. Hence a state average says very little, and the planning has to be done at the district and block level. For higher secondary, about 43 % of the inequality is between the states. Here the gap is national, and states like Bihar, Jharkhand and Uttar Pradesh need extra higher secondary capacity as a whole.

Kerala is an example of what near universal access looks like. Above lower primary, less than 1 % of the children are beyond the norm. But the lower primary distance is still a gap in Idukki and Wayanad, where the hills and scattered settlements make a 1 km walk difficult. The 411 priority areas show the places where a new lower primary school would help the most.

4.2 What I learned about the big data tools

Spark is not always faster. For a small job, the time taken to start Spark and plan its tasks is longer than the job itself, and in such cases plain Python is faster. Spark was useful only when the data was large, and mainly when the whole of India had to be processed. Running India state by state, with each finished state being saved, meant that a failure cost only minutes and not hours.

The join strategy mattered more than the number of cores. When one side is small, like the schools of a state, broadcasting it with a spatial index was found to be three to four times faster than spatial partitioning, whereas the KDB-tree partitioning did not finish at all for Uttar Pradesh. It was also found that Sedona plans a fast broadcast join only for some forms of the distance filter, hence the query plan had to be read in order to check it.

On my 16 GB laptop, memory was the real limit. Twelve Spark workers ran out of RAM, hence only six workers were used. The benchmark also showed that more cores stop helping once the fast cores of the CPU are used. Storing everything as Parquet in bronze, silver and gold layers kept the files small, fast to read and easy to check at every step.

4.3 Limitations

- The first plan was wider than the present report. Along with schools, it also included the health facilities for the whole of India and the fire and rescue stations for Kerala. I collected their locations (228,986 health facilities and 129 fire stations), but E2SFCA also needs the capacity of each facility, such as beds and doctors for health, or staff and vehicles for fire stations. This data is not openly published. It has to be requested from the government departments, and getting a reply takes longer than the time I had for this project. So I limited the study to schools, where capacity could be estimated from open data.
- Distances are straight lines, but the RTE norm is a walking distance. A real walk is always longer, so the share of children beyond the norm is a minimum. The higher secondary 7 km norm is in practice travelled by bus or bicycle, but the straight line is still a lower bound for any route.
- The UDISE school file was taken in January 2022 and most likely shows 2020-21. The file itself does not state its school year. Newer UDISE+ years are published only as reports and tables without school locations, and from 2022-23 they count individual students, so the years are not directly comparable.
- No source gives the number of seats in a school, so capacity was estimated from teachers and the pupil-teacher norms. The current enrolment was used as a check, and both runs ranked districts in a similar way.
- The ages within each WorldPop 5-year band are split evenly. The Sprague method, which was applied to raster data by IIEP-UNESCO (Gagnon and Vargas Mesa, 2021), is the recommended method. But it needs more age bands, and it can also give negative counts in the small 100 m cells. Hence it is not used here. In Kerala, the even split was found to be within 4 % of the real enrolment.
- The UDISE list is incomplete in some places. Delhi, Andhra Pradesh higher secondary, and Dadra and Nagar Haveli and Daman and Diu are flagged.

- In West Bengal almost no private schools are listed, so the Government and Aided share is close to 100 % and places per child there are understated.
- About 1.7 % of schools still share one point with four or more others, usually a village centre, so their distances are less exact.
- Seven Kerala local bodies have no OpenStreetMap boundary, and each 100 m cell is given whole to the area that holds its centre.

4.4 Conclusion and future work

This study measured school access for every 100 m cell in India and found two separate problems: distance for primary children in remote areas, and a lack of places in crowded areas, worst at higher secondary. About 12.4 million primary children live more than 1 km from a primary school, and most of the inequality lies inside states. Spark with Sedona made it possible to process 140 million cells and over a million schools on one laptop.

The next steps are to use walking distances along the road network, to update the schools when newer school-level UDISE+ data becomes open, and to test the Sprague age split. A useful policy scenario would be to ask whether the present Government and Aided schools could take all children, including those now in private schools. The same pipeline can also be used for health centres and fire stations, which were left out of this study.

References

Agafonkin, V. (2023). *Leaflet 1.9.4: an open-source JavaScript library for interactive maps*. https://leafletjs.com

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300.

Dai, D. (2010). Black residential segregation, disparities in spatial access to health care facilities, and late-stage breast cancer diagnosis in metropolitan Detroit. *Health & Place*, 16(5), 1038–1052.

Gagnon, A. A., & Vargas Mesa, G. (2021). *Estimating school-age populations by applying Sprague multipliers to raster data*. IIEP Technical Notes, No. 1. IIEP-UNESCO, Paris.

Getis, A., & Ord, J. K. (1992). The analysis of spatial association by use of distance statistics. *Geographical Analysis*, 24(3), 189–206.

Government of India (2010). *The Right of Children to Free and Compulsory Education Rules, 2010* (Model Rules). Ministry of Human Resource Development, New Delhi.

Government of Kerala (2026a). *Higher Secondary Centralised Allotment Process (HSCAP): school and course list 2026*. General Education Department. https://hscap.kerala.gov.in/

Government of Kerala (2026b). *VHSE prospectus 2026 and Annexure 1, school course list*. VHSCAP, General Education Department. https://www.vhscap.kerala.gov.in/

Han, J., Pei, J., & Yin, Y. (2000). Mining frequent patterns without candidate generation. *Proceedings of ACM SIGMOD 2000*, 1–12.

India Data Portal (2022). *Basic details of schools (UDISE)* [dataset]. Extracted from the UDISE+ school search (https://src.udiseplus.gov.in/locateSchool/schoolSearch), retrieved 12 January 2022.

KITE (2026). *Sametham: school details portal, 2026-27*. Kerala Infrastructure and Technology for Education. https://sametham.kite.kerala.gov.in/

Luo, W., & Qi, Y. (2009). An enhanced two-step floating catchment area (E2SFCA) method for measuring spatial accessibility to primary care physicians. *Health & Place*, 15(4), 1100–1107.

Luo, W., & Wang, F. (2003). Measures of spatial accessibility to health care in a GIS environment: synthesis and a case study in the Chicago region. *Environment and Planning B: Planning and Design*, 30(6), 865–884.

Ministry of Education, Government of India (2022). *Samagra Shiksha: Framework for Implementation*. New Delhi. https://samagra.education.gov.in/docs/ss_implementation.pdf

OpenStreetMap contributors (2026). OpenStreetMap data for India and Kerala, extracted from Geofabrik, 7 November 2026. Open Database License (ODbL).

Rousseeuw, P. J. (1987). Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65.

Runfola, D., et al. (2020). geoBoundaries: a global database of political administrative boundaries. *PLoS ONE*, 15(4), e0231866.

Theil, H. (1967). *Economics and Information Theory*. North-Holland, Amsterdam.

WorldPop (2025). Global population and age-sex structures, release R2025A, 100 m, India 2026. University of Southampton. doi:10.5258/SOTON/WP00841.

Yu, J., Wu, J., & Sarwat, M. (2015). GeoSpark: a cluster computing framework for processing large-scale spatial data. *Proceedings of ACM SIGSPATIAL 2015*, 1–4.

Zaharia, M., et al. (2016). Apache Spark: a unified engine for big data processing. *Communications of the ACM*, 59(11), 56–65.